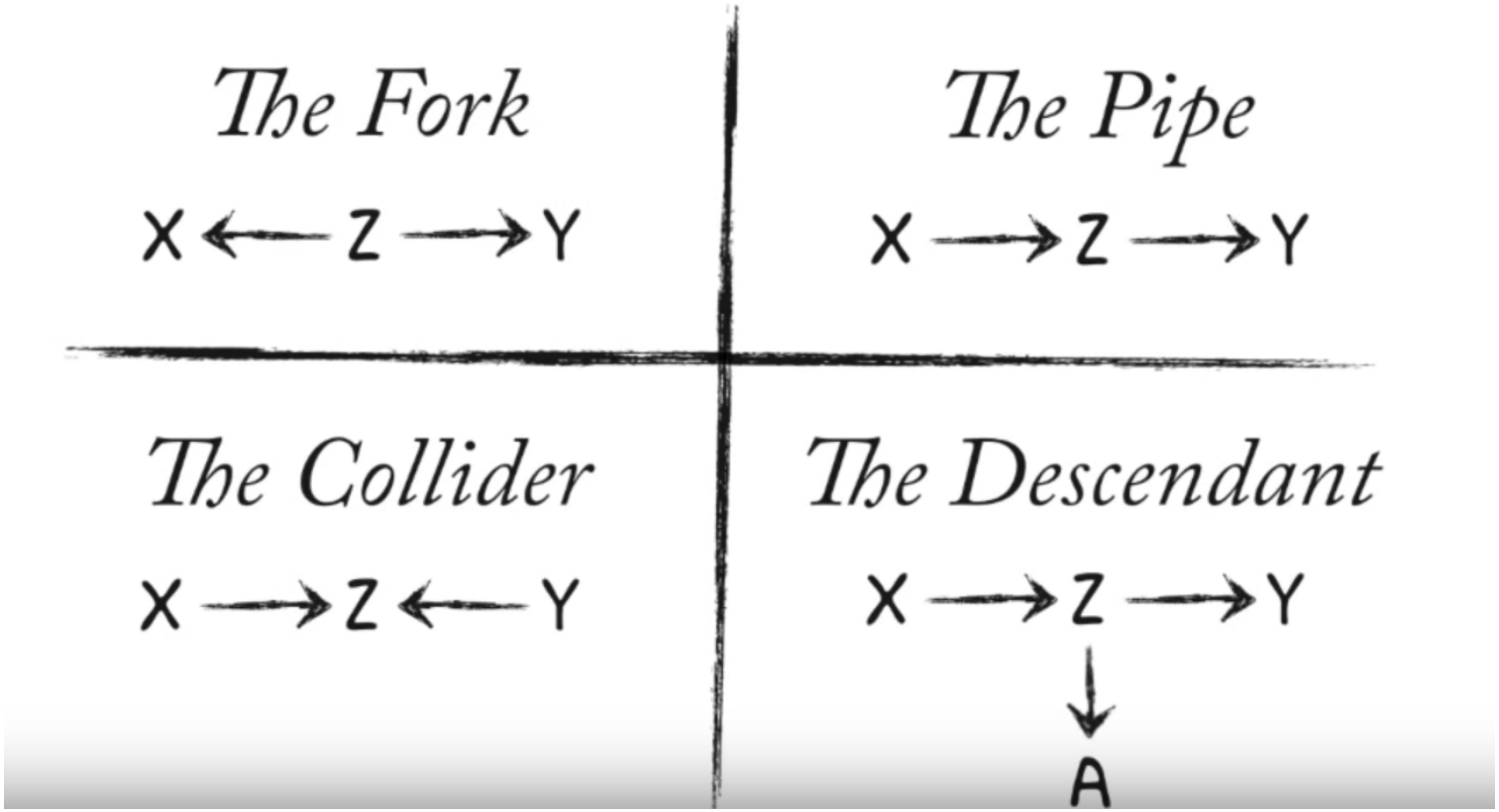
Lecture 16: Reminder: the causal cutlery



Reminder: the causal cutlery

The Fork



Open unless you condition on Z

The Pipe



Open unless you condition on Z

The Collider



Closed until you condition on Z

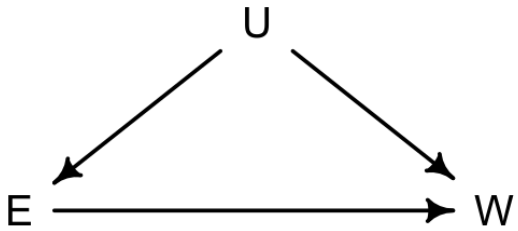
The Descendant



Conditioning on A is like conditioning on Z

Closing the "pipe"

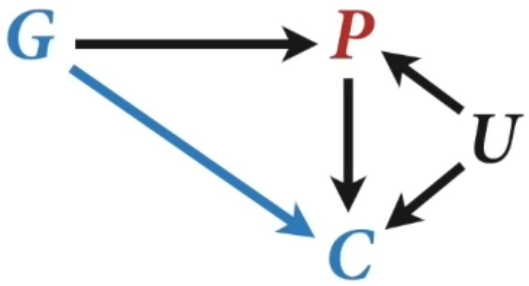
Or the fork



- Two paths from E to W:
  1. $E \rightarrow W$ and
  2. $E \leftarrow U \rightarrow W$
- The second path can be closed by conditioning on U.

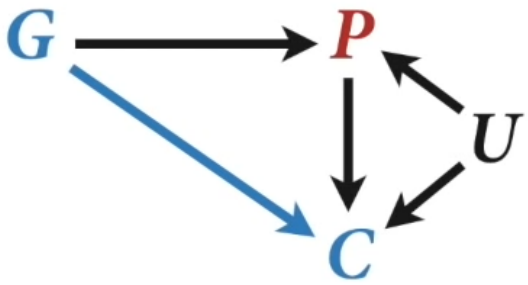


Good and bad controls I



- This is where we left off last week.

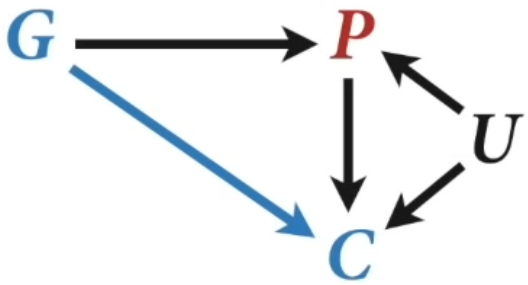
Good and bad controls I



- Three paths from $G \to C$:
  1. $G \to C$,
  2. $G \to P \to C$ and
  3. $G \to P \leftarrow U \to C$.
- Conditioning on P closes the second path but opens the third one.

The "haunted" DAG

- Unmeasured variables can also creates colliders.



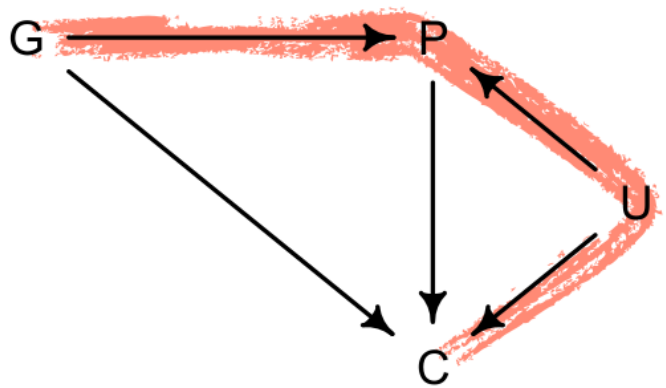
- Simulating haunting

```
n <- 500 # number of GP-P-child triples
b_GP <- 1 # direct effect of G on P
b_GC <- 0 # direct effect of G on C
b_PC <- 1 # direct effect of P on C
b_U <- 2 # direct effect of U on P and C
```

Simulated haunting

```
set.seed(1)
U <- 2*rbern(n, 0.5)-1 ; G <- rnorm(n)
P <- rnorm(n, b_PC*P + b_U*U)
C <- rnorm(n, b_PC*P + b_GC*G + b_U*U)
d <- data.frame(C = C, P = P, G = G, U = U)
```

```
m6.11 <- quap(alist(
  C ~ dnorm(mu, sigma),
  mu <- a + b_PC*P+b_GC*G,
  a ~ dnorm(0,1),
  c(b_PB, b_GC) ~ dnorm(0,1),
  sigma ~ dexp(1)
), data = d)
precis(m6.11)
        mean   sd  5.5% 94.5%
a      -0.12 0.10 -0.28  0.04
b_PC    1.79 0.04  1.72  1.86
b_GC   -0.84 0.11 -1.01 -0.67
sigma   1.41 0.07  1.30  1.52
```

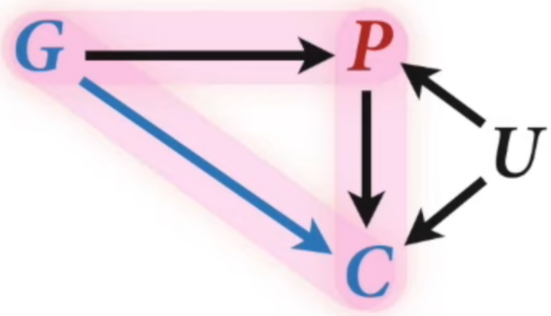


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Bias in the *G*, *P*, *C* example

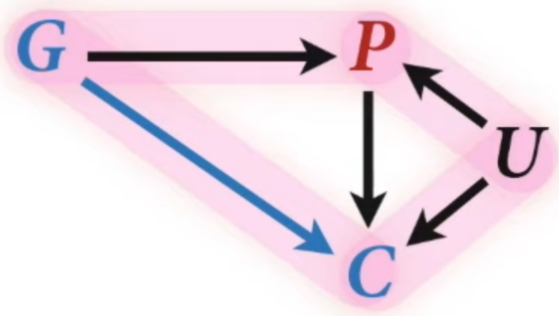
Left with only bad choices

→ One can estimate the *total* causal effect of *G* on *C*.



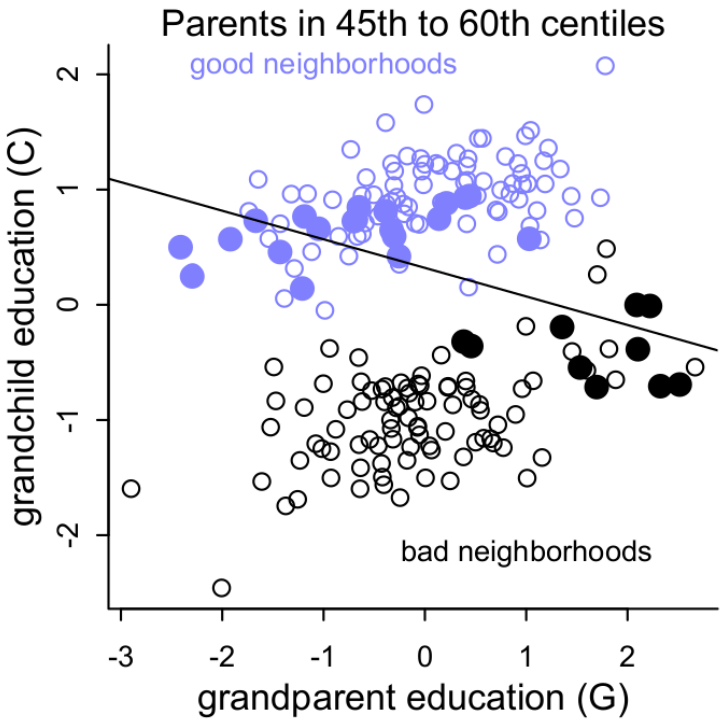
$$
\begin{aligned}
C_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_G G_i
\end{aligned}
$$

→ One *cannot* estimate the *direct* causal effect of *G* on *C*.



$$
\begin{aligned}
C_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_G G_i + \beta_P P_i
\end{aligned}
$$

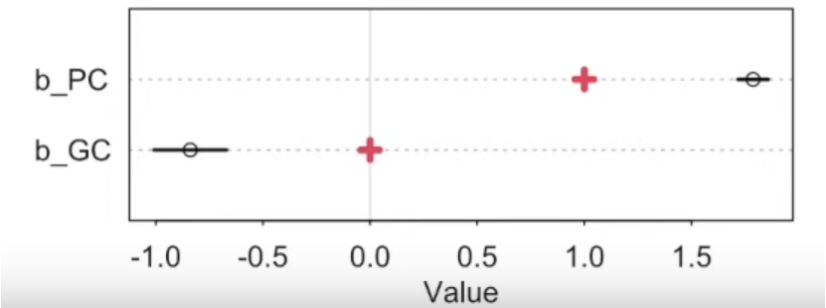
Bias in the *G*, *P*, *C* example

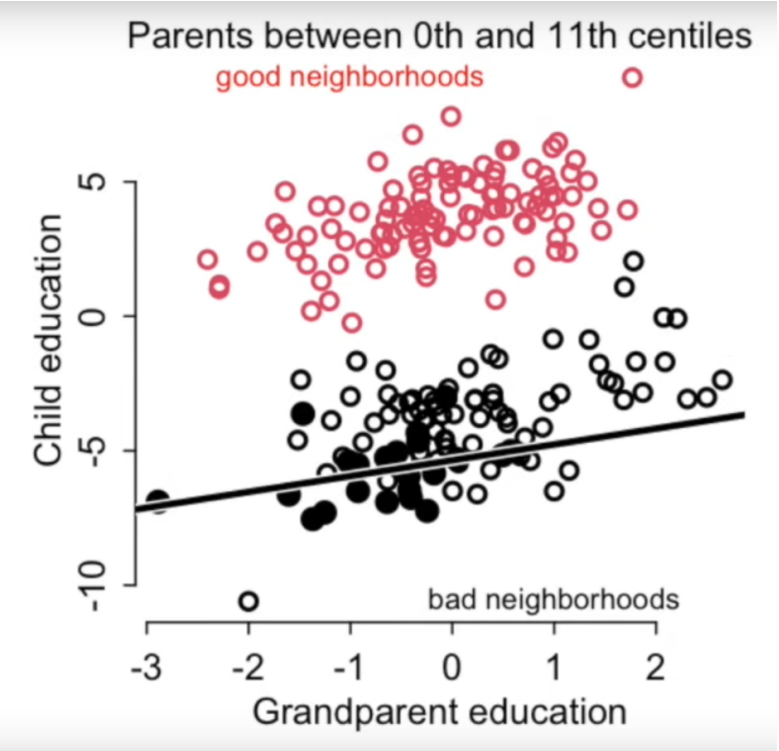


- Consider the *P* in the $45 - 60$th centile of education.
- *P* in good neighbourhoods "must" have less educated *G*.
- *P* in bad neighbourhoods "must" have more educated *G*.
- Otherwise they would not be in the same quantile.

This is an example of collider bias

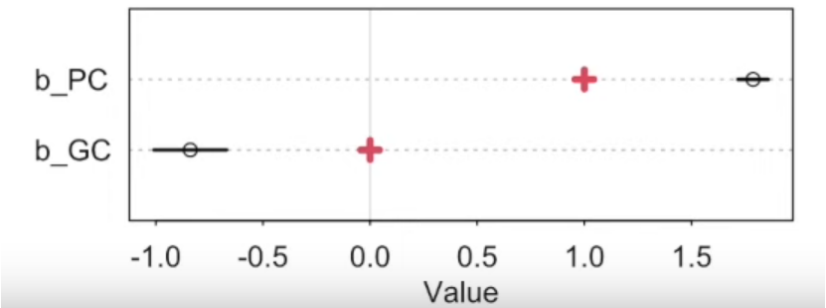
- Stratify by parent education (centile).
- This is a collider.
- Two ways for parents to attain their education: from *G* or from *U*.

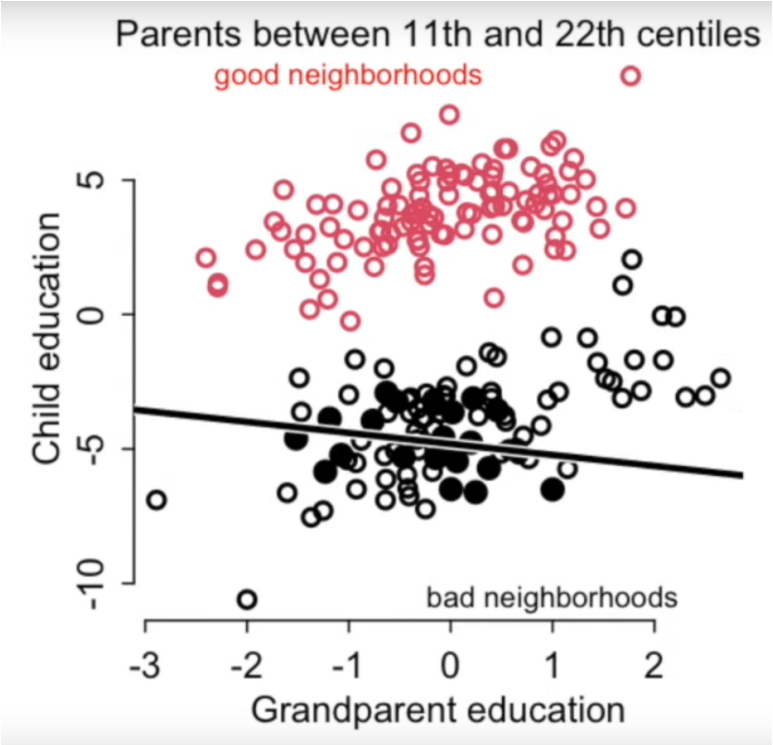




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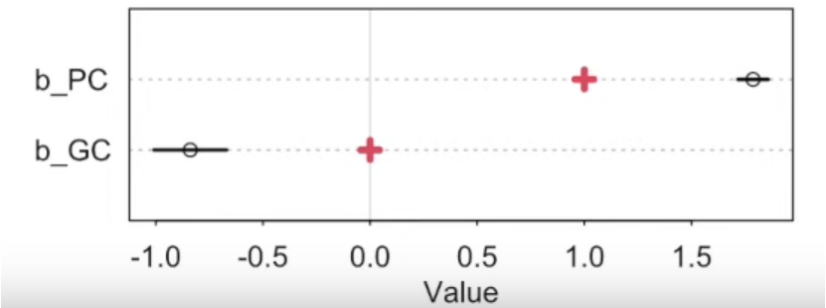
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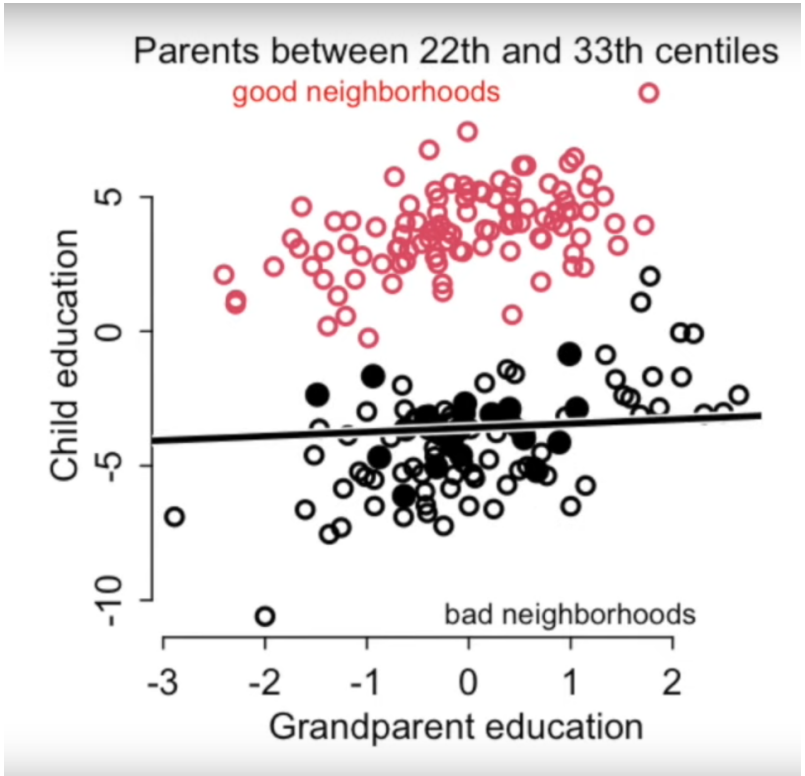




This is an example of collider bias

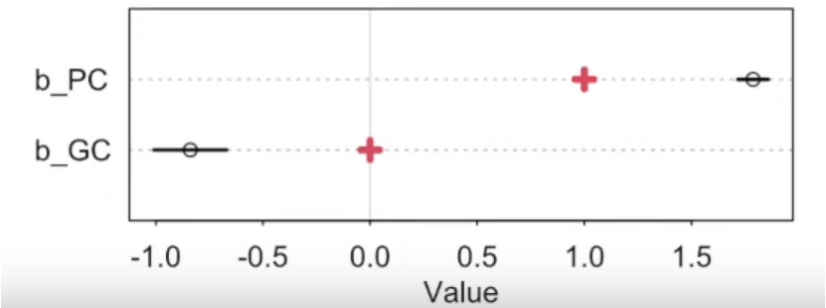
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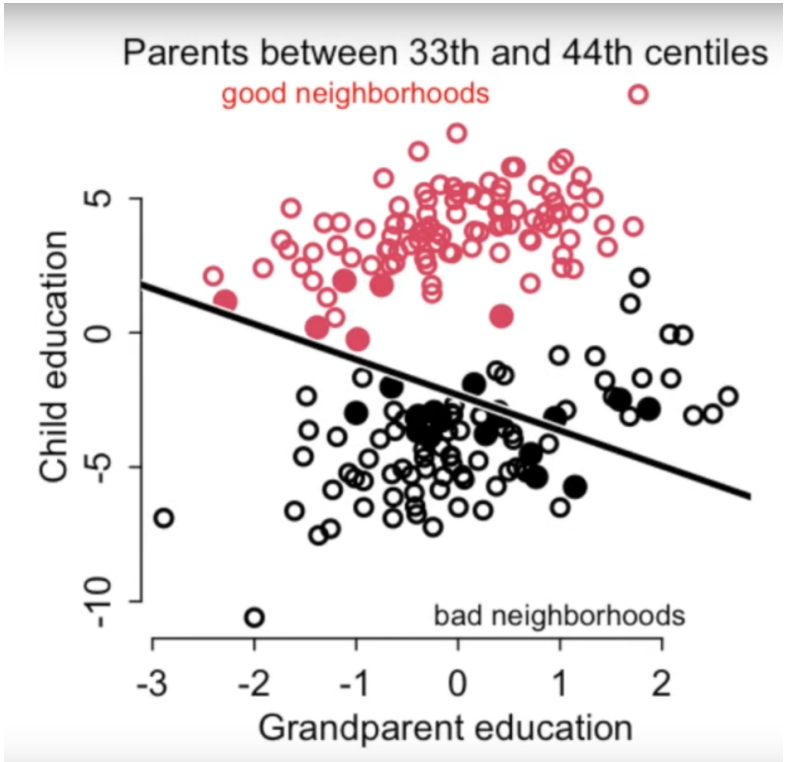




This is an example of collider bias

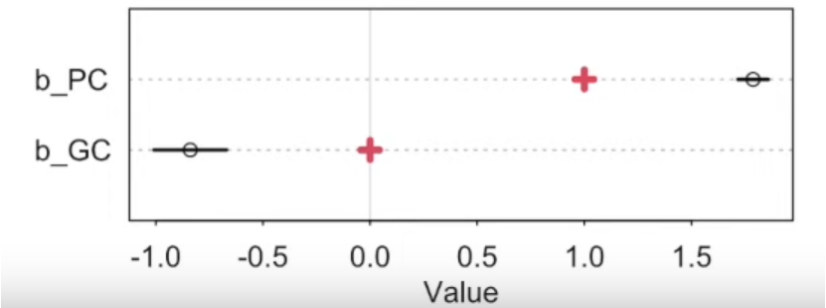
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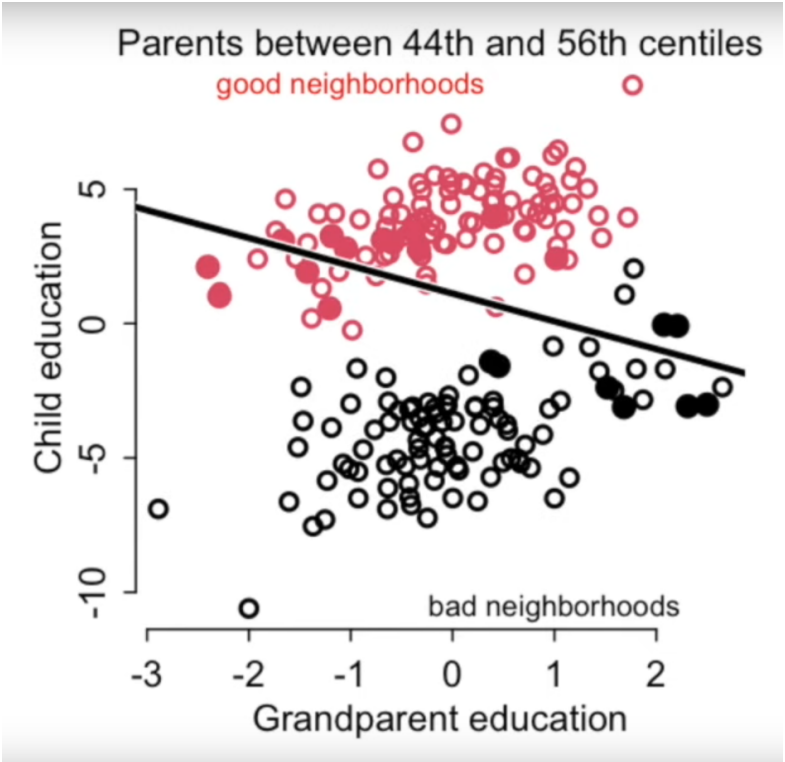




This is an example of collider bias

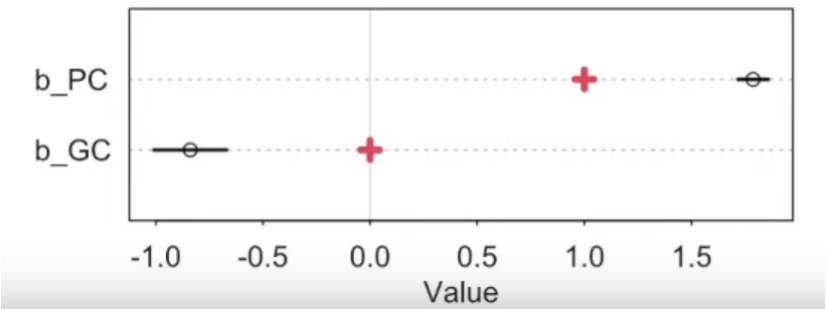
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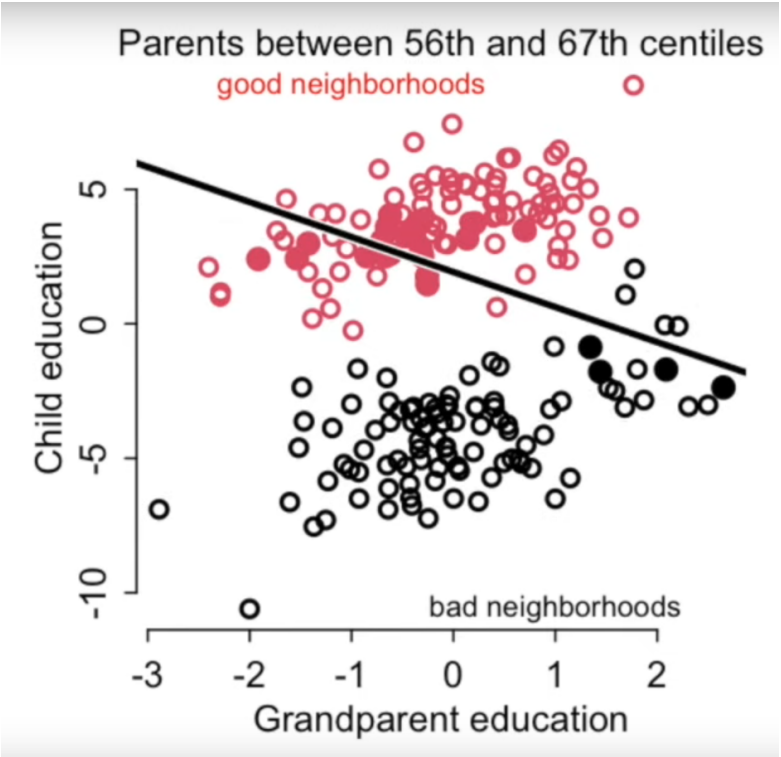




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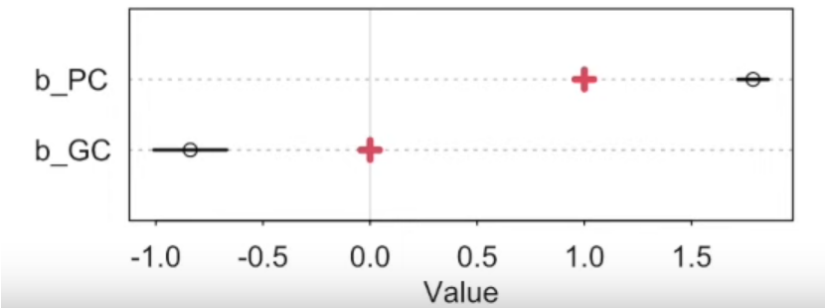
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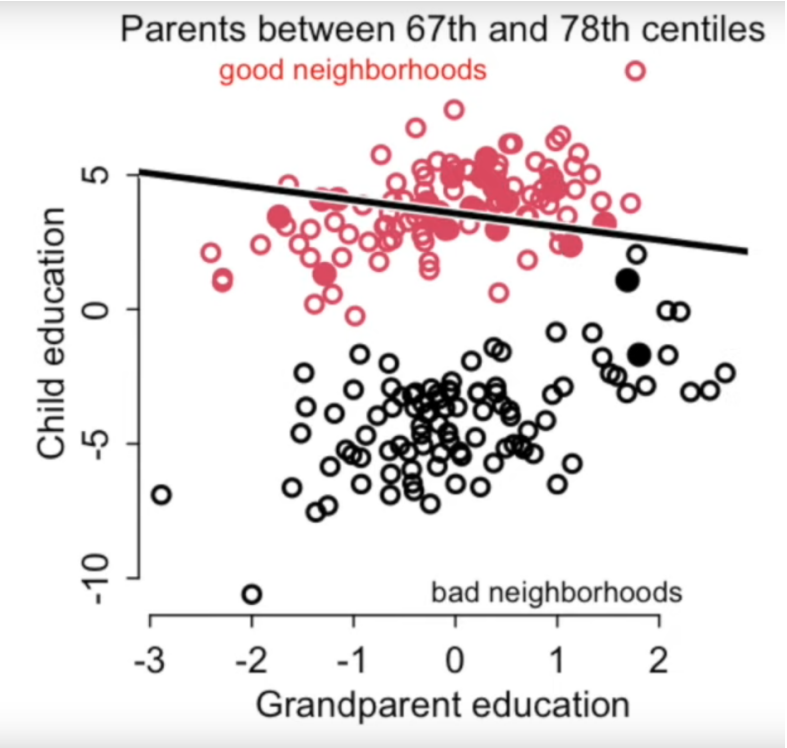




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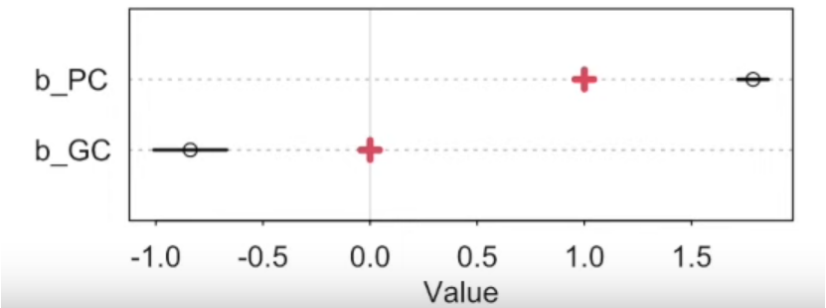
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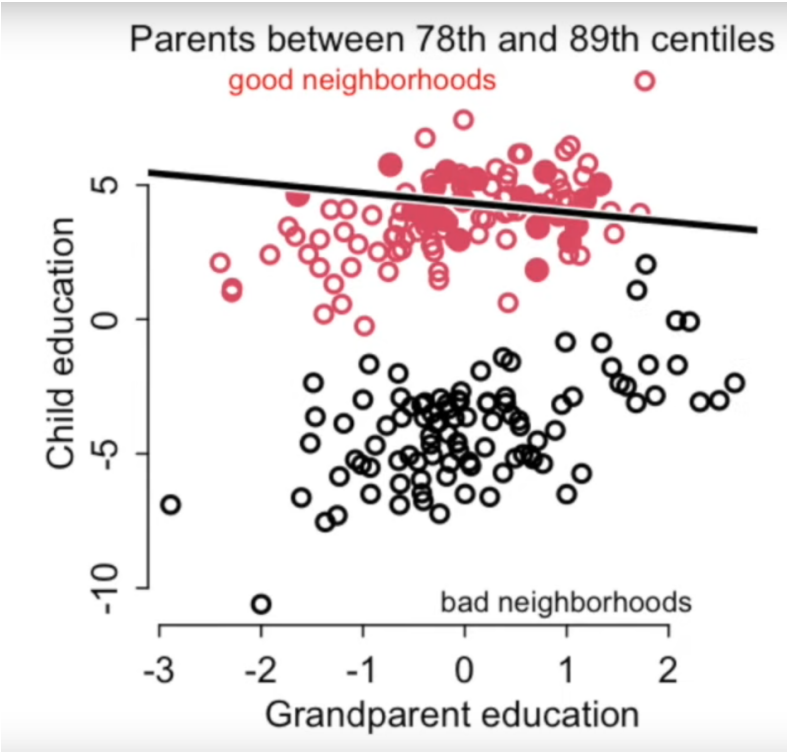




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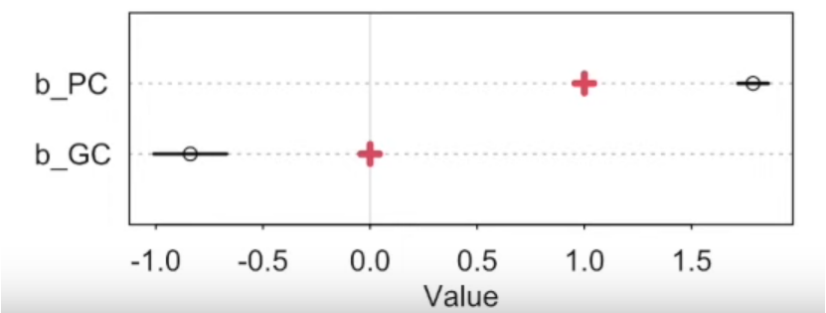
- Stratify by parent education (centile).
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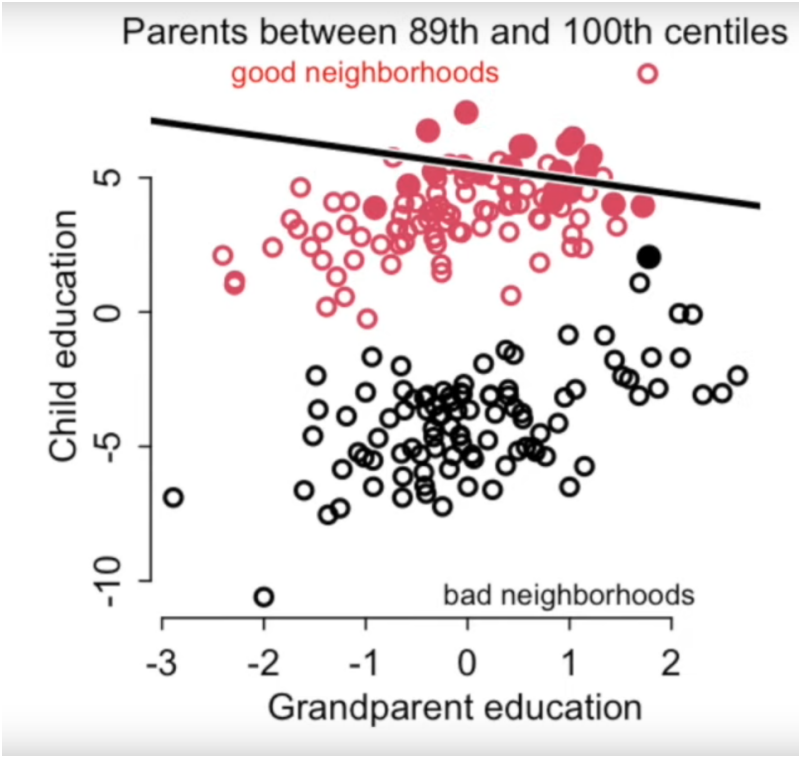




This is an example of collider bias

- Stratify by parent education (centile).
- This is a collider.
- Two ways for parents to attain their education: from *G* or from *U*.





Stratifying by one variable means that for each level/value range of that variable, you look at the association among other variables. Within its levels rather than in the total sample.

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Back to basics: from theory to estimates

- Our job is to:
  - Clearly state assumptions
  - Deduce implications
  - Test implications
- Avoid being clever: neither transparent nor reliable.
- Given a causal model (doesn't have to a DAG), use logic to derive implications.
- Others can use the same logic to verify/challenge your work.

Reminder: the causal cutlery

The Fork



Open unless you condition on Z

The Pipe



Open unless you condition on Z

The Collider



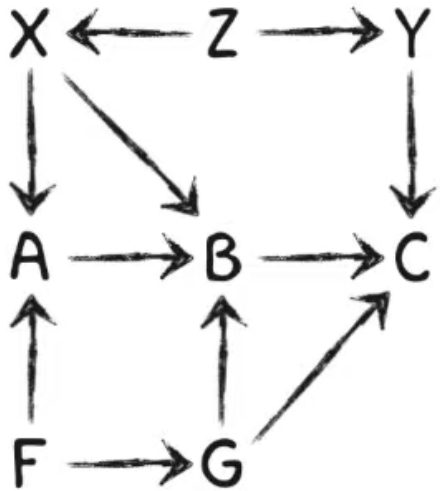
Closed until you condition on Z

The Descendant



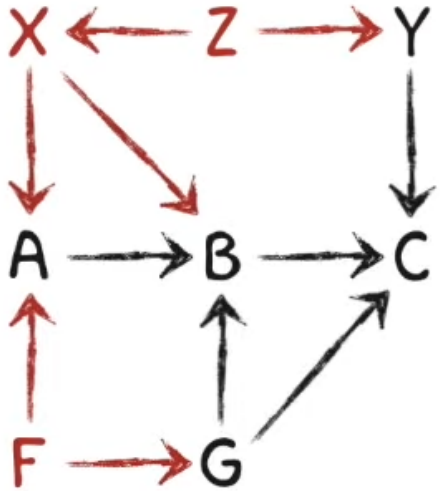
Conditioning on A is like conditioning on Z

Application to real-life graphs



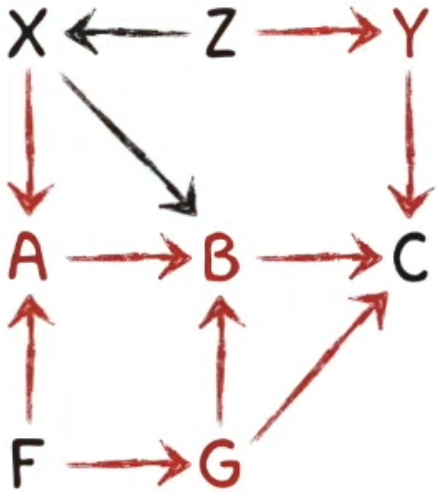
Application to real-life graphs

Forks



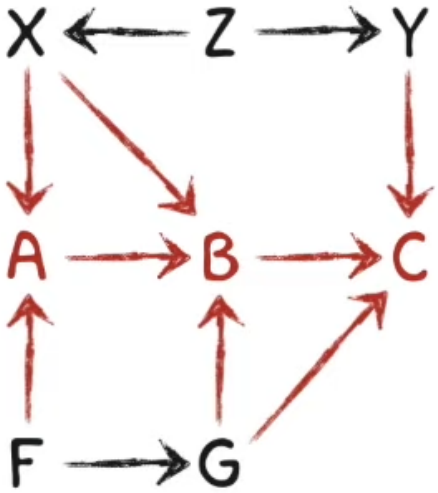
Application to real-life graphs

Pipes



Application to real-life graphs

Colliders

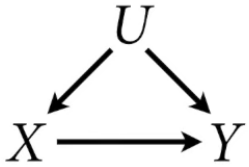


- Some variables (many) are in more than one elementary block
- And we have seen that descendants' role depends on whom they descend from.

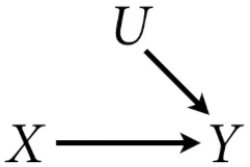
DAG thinking

- In an experiment, we cut causes of the treatment.
- We *randomise* (hopefully)
- So how does causal inference can ever work without randomisation?
- Is there a statistical procedure that mimics randomisation?

Without randomisation



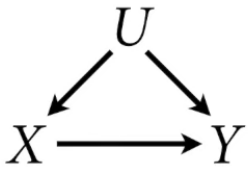
With randomisation



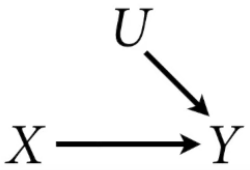
DAG thinking

- Is there a statistical procedure that mimics randomisation?
- The good answer is "sometimes"
- $P(Y|\mathrm{do}(X)) = P(Y|?)$
- $\mathrm{do}(X)$ means intervene on X.
- We are looking at some statistical manipulation (conditioning) that mimics this.
- Can analyse causal model to find answer (if it exists)

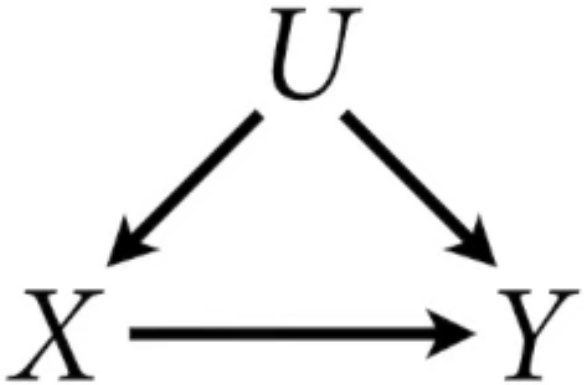
Without randomisation



With randomisation

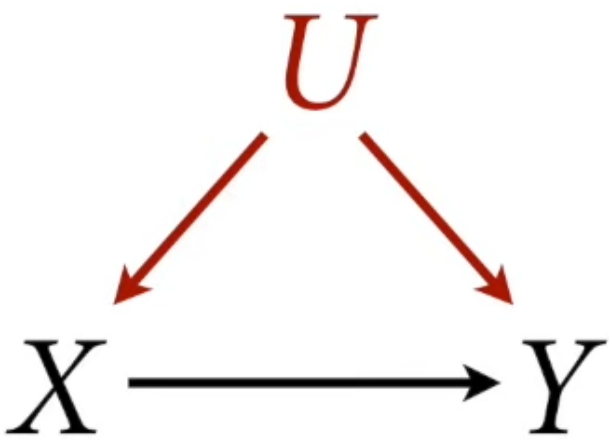


Example: simple confound



Example: simple confound

Conditioning on *U*

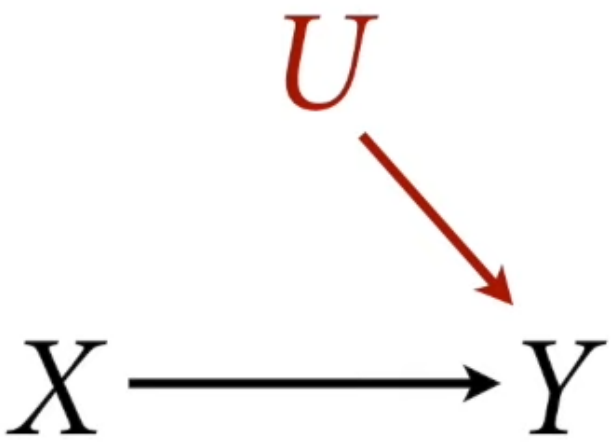


Non-causal path
X <– U –> Y

Close the fork!
Condition on U

Example: simple confound

An intervention experiment on *U*



Non-causal path

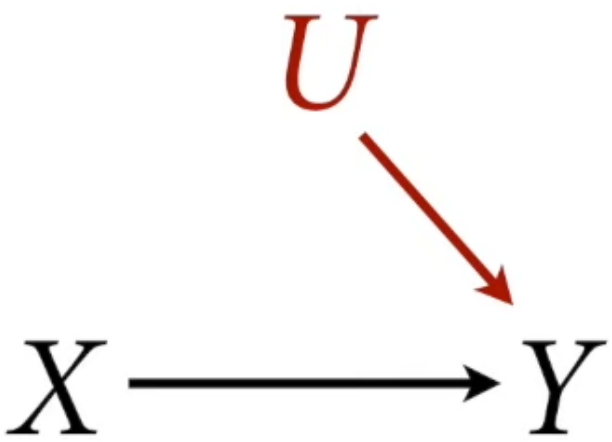
X <– U –> Y

Close the fork!

Condition on U

Example: simple confound

An intervention experiment on U



Non-causal path
X <– U –> Y

Close the fork!
Condition on U

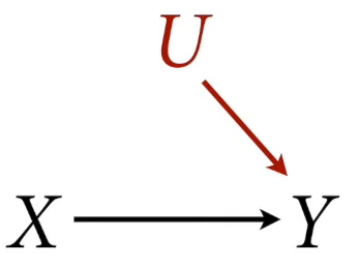
$$P(Y|\mathrm{do}(X)) = \sum_U P(Y|X,U)P(U) = \mathbb{E}_U P(Y|X,U)$$

"The distribution of Y, stratified by X and U, averaged over the distribution of U".

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Example: simple confound

An intervention experiment on U



Non-causal path
X <– U –> Y

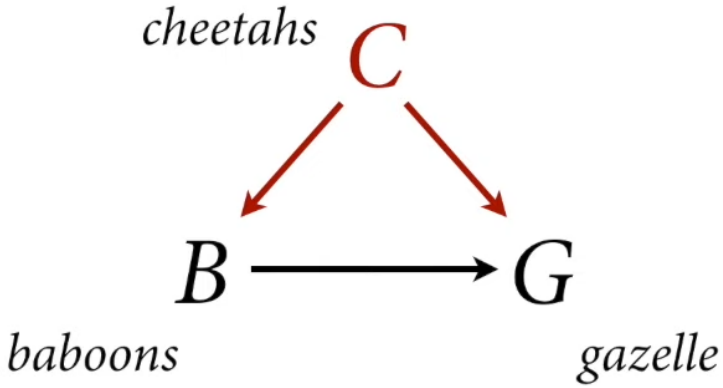
Close the fork!
Condition on U

$$P(Y|\mathrm{do}(X)) = \sum_U P(Y|X,U)P(U) = \mathbb{E}_U P(Y|X,U)$$

"The distribution of Y, stratified by X and U, averaged over the distribution of U".

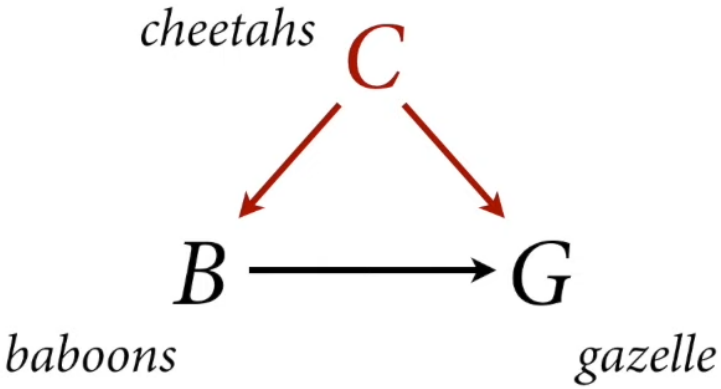
- The causal effect of X on Y is not (in general) the coefficient relating X to Y.
- It is the distribution of Y when we change X, averaged over the distribution of the control variables. Here U.

Marginal effects are the way to think about causal effects



Say you want to estimate the predation pressure of baboons on gazelles.
What is the confound?

Marginal effects are the way to think about causal effects



Causal effects of baboons depends upon distribution of cheetahs. (example of this in Chapter 6 where we simulate causal effects)

Do-calculus

- For DAGs, rules for finding $P(Y|\mathrm{do}(X))$ known as *do-calculus*. Only 3 of them.
- do-calculus says what is possible to say before picking functions.
- Additional assumptions yield additional implications. These could be exploited to get more powerful estimation.
- The do-calculus is both (i) worst case (additional assumptions often allow stronger inference) and (ii) best case (if inference is possible by do-calculus, it is independent on special assumptions) claims

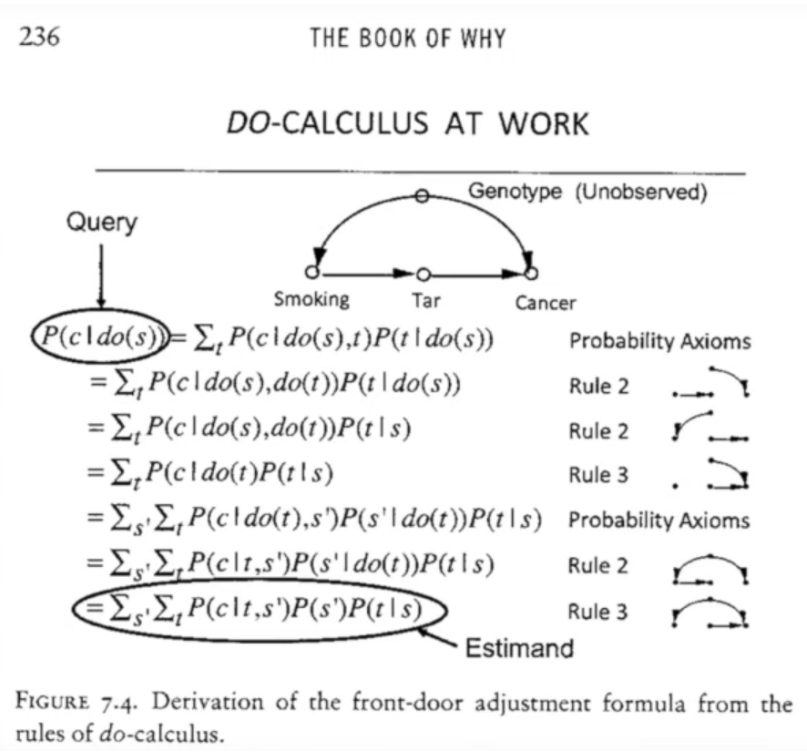
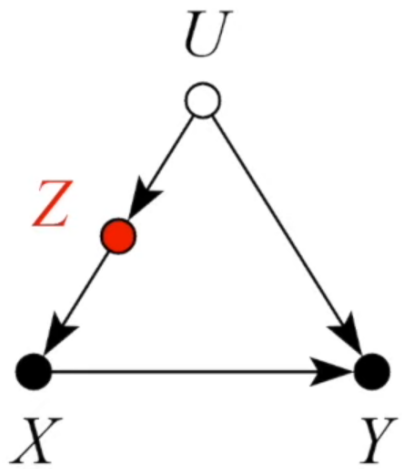


FIGURE 7.4. Derivation of the front-door adjustment formula from the rules of *do*-calculus.

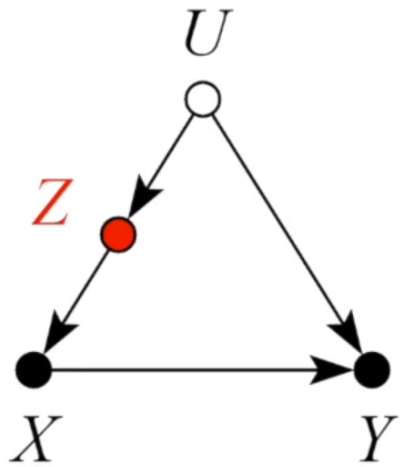
Backdoor criterion

- Very useful implication of the do-calculus is the *Backdoor Criterion*.
- The backdoor criterion is a shortcut to applying the do-calculus rules. The do-calculus is bigger than backdoor criterion, but the BC is extremely useful.
- Helps analysing graphs to know which variables to stratify by.
- Also inspires strategies for research design that yield valid estimates.



Backdoor criterion

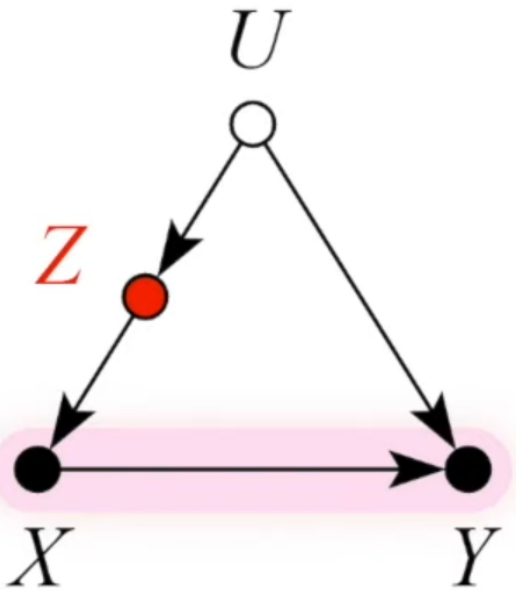
- backdoor criterion: rule to find a set of variables to stratify (condition) by to yield $P(Y|\mathrm{do}(X))$. In three steps:
  1. Identify all *path* connecting the treatment (X) to the outcome (Y).
  2. Paths with arrows entering X are backdoor paths (non causal paths).
  3. Find adjustment set that closes/blocks all backdoor paths.

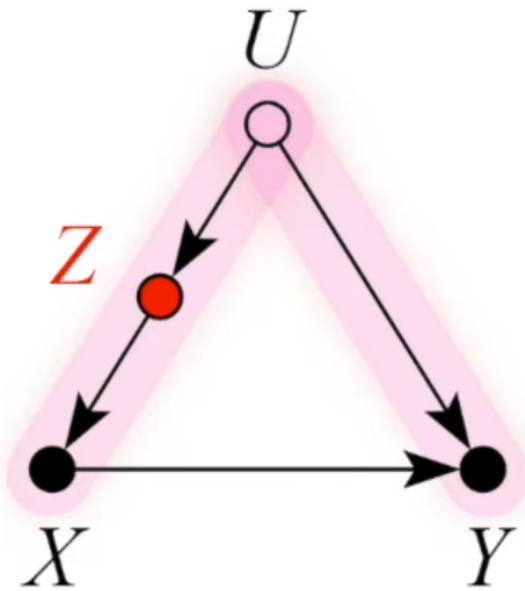


Backdoor criterion example

Work through

1. Identify all *path* connecting the treatment (X) to the outcome (Y).

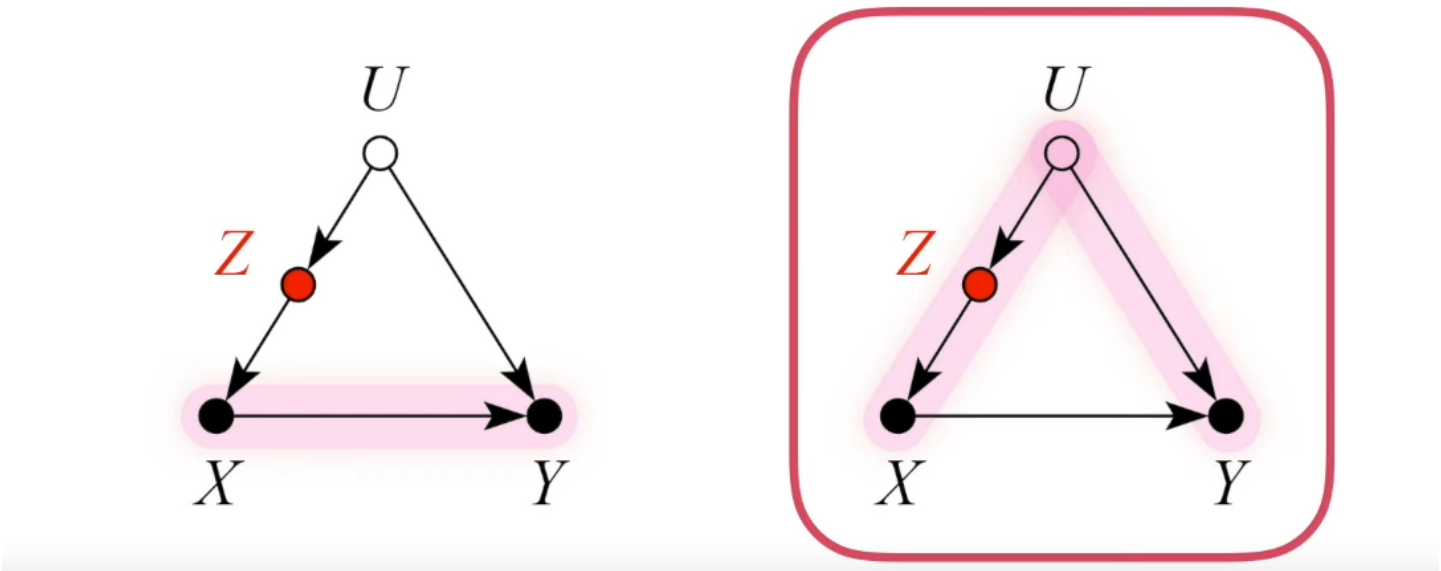




Backdoor criterion example

Work through

2. Paths with arrows entering X are backdoor paths (non causal paths).

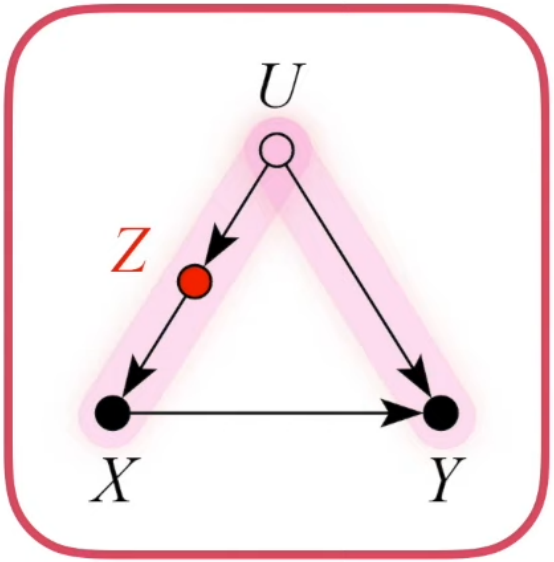


Backdoor criterion example

Work through

3. Paths with arrows entering X are backdoor paths (non causal paths).

Block the pipe: $X \perp\!\!\!\perp U \mid Z$



Backdoor criterion example

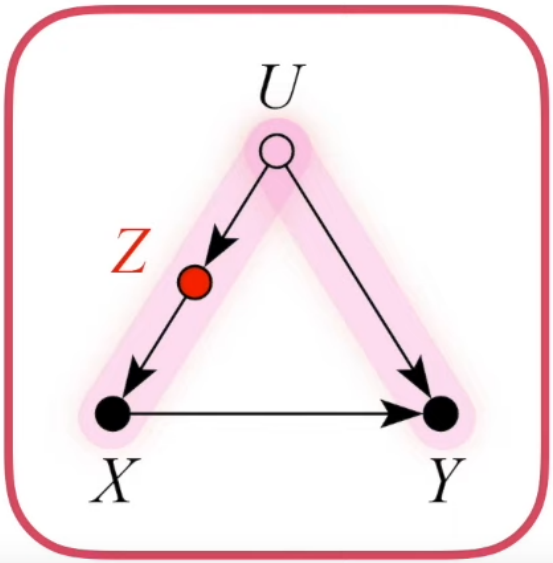
Work through

3. Paths with arrows entering X are backdoor paths (non causal paths).

Block the pipe: $X \perp\!\!\!\perp U \mid Z$

$$P(Y \mid \text{do}(X)) = \sum_U P(Y \mid X, Z) P(Z)$$

$$Y_i \sim \text{Normal}(\mu_i, \sigma)$$
$$\mu_i = \alpha + \beta_X X_i + \beta_Z Z_i$$



For tomorrow

Some homework

List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.

